A heralded controlled-phase gate from radial return in an independent-field DBI model

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Abstract

A common radial source, finite autonomous evolution and a rank-one radial herald define a controlled-phase gate in a specified independent-field Dirac–Born–Infeld (DBI) half-line model. At the frozen dimensionless time 107.61733846478977, a diagonal successful Kraus operator lies within 0.18972055982017294 of a unitary locally equivalent to controlled-Z. Retained full-half-line residual certificates give success probability at least 0.6565527711781338 for every logical input and conditional squared fidelity at least 0.96400610918152, including arbitrary reference systems. A stated source, readout, timing and local-phase error box retains success at least 0.6351321846638459 and conditional squared fidelity at least 0.9587714318247369. The calculation uses a complete success/failure instrument and an elementary sharp norm-to-conditional-fidelity inequality. The protocol assumes exact code preservation and access to the specified radial preparation, herald and local phase gates; their apparatus costs are not certified.

1 The fixed model and the gate construction

We adopt the dimensionless independent-field DBI radial model and invariant four-state logical code of the retained continuum calculation [1, 2]. Its matter prescription is a sum of independent field Lagrangians, $\sum_j P(X_j)$; it is not the cross-coupled prescription $P(\sum_j X_j)$. The model describes specified global zero-mode sectors, not spatially separated laboratory systems. The present result is a new return-herald protocol within this fixed quantization, rather than a claim that this quantization is uniquely implied by a gravitational action.

Let $\mathcal{H}_r = L^2((0,\infty), dy)$ and $s = y^2$. On the self-adjoint half-line realization inherited from the reference oscillator, define

$$A_{20} = -\partial_y^2 + y^2 + 20y^{-2}, \tag{1}$$

$$A_b = A_{20} - F_b(y^2), \qquad F_b(s) = \sum_{q\in\mathcal{Q}_b} (\sqrt{s^2+2q} - s), \tag{2}$$

$$q_{\rm bg} = \frac{121}{729}, \quad q_{\rm lo} = \frac{121}{1458}, \quad q_{\rm hi} = \frac{4741}{1458},$$
$$\mathcal{Q}_{00} = (q_{\rm bg}, q_{\rm lo}, q_{\rm lo}), \quad \mathcal{Q}_{01} = \mathcal{Q}_{10} = (q_{\rm bg}, q_{\rm lo}, q_{\rm hi}),$$
$$\mathcal{Q}_{11} = (q_{\rm bg}, q_{\rm hi}, q_{\rm hi}).$$

All fields are at the fixed finite nonlinear parameter $\theta = 1$. For $s \geq 0$, $0 \leq F_b(s) \leq \sum_{q\in\mathcal{Q}_b} \sqrt{2q}$, so F_b is bounded and $D(A_b) = D(A_{20})$. Our dust-time convention is explicitly $U_b(\tau) = e^{+iA_b\tau}$. Changing this sign would change the frozen phases; it is not a free convention within the certificate.

The normalized common radial source is

$$\psi_5(y) = \sqrt{\frac{2}{\Gamma(11/2)}}\, y^5 e^{-y^2/2}, \qquad A_{20}\psi_5 = 11\psi_5. \tag{3}$$

Prepare it independently of an arbitrary two-qubit logical state, possibly entangled with an arbitrary reference. Evolve with $\sum_b |b\rangle\langle b| \otimes U_b(\tau)$, then measure the radial outcomes

$$P = |\psi_5\rangle\langle\psi_5|, \qquad Q = I - P, \tag{4}$$

keeping a flag for both outcomes. The successful outcome resets the radial state exactly and induces the contraction

$$K = \mathrm{diag}(k_{00}, k_{01}, k_{01}, k_{11}), \qquad k_b = \langle\psi_5|U_b(\tau)|\psi_5\rangle. \tag{5}$$

The time and target phases are frozen decimal numbers,

$$\begin{aligned} \tau_* &= 107.61733846478977, \\ \phi_0 &= 0.9177158589349711, \qquad \phi_1 = 0.69393706341036732, \end{aligned} \tag{6}$$

and the target is the exactly unitary matrix

$$D = \mathrm{diag}(e^{i\phi_0}, e^{i\phi_1}, e^{i\phi_1}, -e^{i(2\phi_1-\phi_0)}). \tag{7}$$

Let $\delta = \phi_1 - \phi_0 = -0.22377879552460378$. Apply $C = \mathrm{diag}(1, e^{-i\delta})$ separately to each qubit on success. Then

$$(C \otimes C)D = e^{i\phi_0}\mathrm{CZ}. \tag{8}$$

In the convention $R_z(\alpha) = \mathrm{diag}(e^{-i\alpha/2}, e^{i\alpha/2})$, the corrective angle is $-\delta$, up to global phase. Equation (8) is algebraic: locating a transcendental root exactly is unnecessary. The time was selected using finite-trial diagnostic return phases, then certified at that exact decimal time. No global optimality of this time is claimed.

2 Full-half-line residual control at the new time

The supplied trial data [2] are exact archived binary64 coefficients, not exact continuous eigenpairs. The normalized basis is $\varphi_n = \psi_5 p_n(s)$, with

$$d\mu(s) = \frac{s^{9/2}e^{-s}}{\Gamma(11/2)}\,ds, \qquad A_{20}\varphi_n = (4n+11)\varphi_n. \tag{9}$$

The polynomials are orthonormal for $d\mu$ and obey $p_0 = 1$ and

$$\sqrt{(n+1)(n+11/2)}\,p_{n+1} = (s - 2n - 11/2)p_n - \sqrt{n(n+9/2)}\,p_{n-1}, \tag{10}$$

where the last term is absent at $n = 0$. For each branch take eight trial functions in the first sixteen basis vectors:

$$f_{bj} = \sum_{n=0}^{15} v_{bnj}\varphi_n, \qquad \widehat{\psi}_b(t) = \sum_{j=0}^{7} c_{bj}e^{i\lambda_{bj}t}f_{bj}. \tag{11}$$

All v, c, λ are in the supplied JSON; exact orthogonality of the trial eigenvectors is unnecessary. For $p_{bj} = \sum_n v_{bnj}p_n$ and $h_{bj} = \sum_n (4n + 11 - \lambda_{bj})v_{bnj}p_n$, the true residual norm is

$$r_{bj}^2 = \|(A_b - \lambda_{bj})f_{bj}\|^2 = \int_0^\infty |h_{bj}(s) - F_b(s)p_{bj}(s)|^2 d\mu(s). \tag{12}$$

The retained certificates [3] bound this entire integral; the half-line operator is not replaced by a spectral cutoff.

Their enclosure algorithm uses interval Taylor bounds on cells in $[0, 128]$. Writing $r(s) = h(s) - F(s)p(s)$, a cell of midpoint m and half-width h_c satisfies

$$\sup_{\text{cell}} |r| \le |r(m)| + h_c|r'(m)| + \tfrac{1}{2}h_c^2 \sup_{\text{cell}} |r''|. \tag{13}$$

The square times the cell width and the maximum gamma density bounds its integral. The density maximum is checked at the endpoints and its sole stationary point $s = 9/2$; derivatives use the actual square roots in (2). For the tail choose a nonnegative-coefficient polynomial P_+ bounding $|h| + F_{\max}|p|$, with degree $d \le 15$. For $s \ge R = 128$, $P_+(s) \le P_+(R)(s/R)^d$. Together with $\log(s/R) \le (s - R)/R$, this yields

$$\int_R^\infty |r|^2 d\mu \le \frac{R^{9/2} e^{-R} P_+(R)^2}{\Gamma(11/2)[1 - (9/2 + 2d)/R]}. \tag{14}$$

The denominator is positive. Saved cell and tail bounds were rounded upward to binary64; replay treats them as exact rational upper bounds. The original interval producer is included unchanged as an algorithm reference; default replay recombines its saved results without regenerating cells.

Let $\epsilon_{0b} = \left\| \psi_5 - \sum_j c_{bj} f_{bj} \right\|$, evaluated from all sixteen exact basis coefficients rather than set to zero by truncation. The spectral theorem and Duhamel's formula give, for every real t,

$$\left\| U_b(t)\psi_5 - \widehat{\psi}_b(t) \right\| \le \epsilon_{0b} + |t| \sum_j |c_{bj}| r_{bj} =: \epsilon_b(t). \tag{15}$$

Thus the residual norms remain usable at the new time (6); an old finite-time error or parameter box cannot simply be copied to that time. Projecting onto normalized ψ_5 gives

$$|k_b - \widehat{k}_b| \le \epsilon_b(\tau_*), \qquad \widehat{k}_b = \sum_j v_{b0j} c_{bj} e^{i\lambda_{bj}\tau_*}. \tag{16}$$

Table 1 displays the branch calculations; replay uses full interval values rather than their abbreviated printed forms. Comparison with (7) yields

$$\|K - D\| = \max_b |k_b - D_{bb}| \le \eta_{\text{nom}} = 0.18972055982017294. \tag{17}$$

Table 1: Residual slopes and propagated errors at τ_*, rounded for display.

Branch	$\sum_j \lvert c_j \rvert r_j$	$\epsilon_b(\tau_*)$	$\lvert \widehat{k}_b - D_{bb} \rvert + \epsilon_b$
00	0.001529585	0.164697384	0.164819225
01/10	0.001495096	0.160974031	0.167620070
11	0.001522302	0.163867864	0.189720560

3 Sharp conditional fidelity and the complete instrument

We use squared fidelity: for a pure target $|v\rangle$, $F_{\text{sq}}(\rho, |v\rangle) = \langle v|\rho|v\rangle$; for mixed targets it is squared Uhlmann fidelity. The following elementary geometric estimate is proved here, not presented as new general mathematics.

Lemma 1. *Let D be unitary and $\|K - D\| \le \eta < 1$. For every normalized system–reference vector Ψ, successful application of K has probability at least $(1 - \eta)^2$ and conditional squared fidelity with $(D \otimes I)\Psi$ at least $1 - \eta^2$. The fidelity constant is sharp, including for contractions K.*

Proof. Put $v = (D \otimes I)\Psi$, $x = (K \otimes I)\Psi$ and $t = \|x\|$. Then $\|x - v\| \le \eta$ and $t \ge 1 - \eta > 0$. Expanding the squared distance gives

$$\Re\langle v, x\rangle \ge \frac{1 + t^2 - \eta^2}{2} \ge t\sqrt{1 - \eta^2}, \tag{18}$$

where the second inequality is $(t - \sqrt{1-\eta^2})^2 \ge 0$. Dividing by t and squaring proves the claim. For sharpness let $a = 1 - \eta^2$ and take

$$K = \begin{pmatrix} a & -\eta\sqrt{a} \\ \eta\sqrt{a} & a \end{pmatrix}, \qquad D = I. \tag{19}$$

Then $(K-I)^\dagger(K-I) = \eta^2 I$, $K^\dagger K = aI$, and the conditional squared fidelity for either coordinate vector is a. □

For a mixed logical/reference input, apply the lemma to a purification and discard the purifying system. Monotonicity of squared fidelity gives the same bound for the normalized conditional mixed state. The common radial source must initially be independent of that input; arbitrary radial/logical correlations are outside this preparation class.

To specify all outcomes, let $V|b\rangle = |b\rangle \otimes U_b(\tau_*)|\psi_5\rangle$. The successful and failed operators $(I \otimes P)V$ and $(I \otimes Q)V$ have adjoint products summing to $V^\dagger V = I$. Tracing radial outputs therefore defines a trace-preserving flagged instrument with completely positive branches. The successful branch, including local correction, is

$$\mathcal{S}(\rho) = (C \otimes C)K\rho K^\dagger (C \otimes C)^\dagger. \tag{20}$$

Its probability is the trace of this linear map; normalization is the explicitly conditional step. No failures are omitted from the physical instrument. Since K is a contraction, its successful subnormalized map satisfies

$$\|\mathcal{S} - \mathcal{U}_{\mathrm{CZ}}\|_\diamond \le 2\eta. \tag{21}$$

This bounds the successful branch, not a deterministic CZ on every run. For input $|{+}{+}\rangle$, $\mathrm{CZ}|{+}{+}\rangle$ is maximally entangled. The conditional logical output obeys

$$\mathcal{N}(\rho_+) \ge F_{\mathrm{sq}}(\rho_+, \mathrm{CZ}|{+}{+}\rangle) - \tfrac{1}{2} \ge \tfrac{1}{2} - \eta^2, \tag{22}$$

Indeed, with $H = (|\Phi\rangle\langle\Phi|)^{T_B}$ for the maximally entangled target, $\|H\|_\infty = 1/2$ gives $F_{\mathrm{sq}} = \mathrm{Tr}(H\rho_+^{T_B}) \le \left\|\rho_+^{T_B}\right\|_1 /2 = 1/2 + \mathcal{N}(\rho_+)$. Here $\mathcal{N}(\rho) = (\left\|\rho^{T_B}\right\|_1 - 1)/2$; the argument also covers mixed conditional outputs.

4 A bounded instrument error box

Let the actual normalized radial source and normalized rank-one readout mode each differ from ψ_5 by Hilbert norm at most 10^{-3}. Allow absolute timing error $|\Delta\tau| \le 10^{-3}$ and angle error at most 10^{-4} on each local phase gate. The Hamiltonian parameters and exact invariant code are unchanged. These are explicit control assumptions, not an inferred apparatus specification. For the nominal source, $F_b(s) \le Q_{\max}/s$ and

$$Q_{\max} = q_{\mathrm{bg}} + 2q_{\mathrm{hi}} = \frac{4862}{729}, \qquad \langle y^{-4}\rangle_{\psi_5} = \frac{4}{63}. \tag{23}$$

Using $A_b\psi_5 = (11 - F_b)\psi_5$ and $F_b \ge 0$ gives

$$\|A_b\psi_5\| \le M_1 = \sqrt{121 + \left(\frac{4862}{729}\right)^2 \frac{4}{63}} < 11.127631935535533. \tag{24}$$

Unitarity and the spectral theorem imply $\|[U_b(\Delta\tau) - I]\psi_5\| \le M_1|\Delta\tau|$. A telescoping decomposition compares timing on the nominal source and handles source and readout changes separately by their norms. Thus no graph-norm bound on an arbitrary norm-close perturbed source is needed for this bounded gate estimate. Two local phase errors add at most twice their individual angle bound in operator norm, giving

$$\Delta\eta \le 10^{-3} + 10^{-3} + 10^{-3}M_1 + 2 \times 10^{-4} < 0.013327631935535532. \tag{25}$$

Adding this to the certified branch bound before its final display rounding yields

$$\eta_{\mathrm{rob}} = 0.20304819175570846 < 1. \tag{26}$$

Replay checks this comparison directly and tests positivity before applying success and fidelity formulas. Table 2 gives the conservative lower bounds. Lemma 1 removes an unnecessary denominator $(1 - \eta)^2$ from a cruder infidelity estimate; success is unchanged. These norm-error assumptions do not transfer unbounded-density or density-RMS guarantees to arbitrary perturbed radial states.

Table 2: Certified successful-branch guarantees; failures retain their own flag.

	Nominal	Bounded error box
Success	≥ 0.6565527711781338	≥ 0.6351321846638459
Conditional F_{sq}, any reference	≥ 0.96400610918152	≥ 0.9587714318247369
Conditional $\mathcal{N}$, input $++$	≥ 0.4640061091815201	≥ 0.45877143182473695

5 Reproduction, scope and provenance

The packet supplies the exact archived trial and three complete residual partitions with hashes. Running `python replay.py` requires only local Python packages and packet files, with no network or old project paths. It verifies the tangent identity and sharp contraction algebra, input hashes, exact rational sums of saved residual cells and analytic tails, and the initial mismatch from all sixteen basis coefficients. It then recomputes the frozen-time amplitudes, operator errors, robust box and sharper guarantees with 110-digit outward interval arithmetic. The unchanged interval producer is included only as a minimal algorithm reference. The underlying cellwise integrand bounds remain inherited certificates, not newly regenerated evidence. This boundary between retained continuum control and freshly replayed consequences is part of the claim.

The protocol is a heralded entangling operation in the specified invariant model. It does not establish a deterministic gate, universal loading/readout interface, spatial locality, a nonhomogeneous truncation theorem, global time optimality or a quantum-gravity completion. Radial preparation/readout and local phases remain idealized controls with unquantified implementation costs. The general lemma is an elementary geometric estimate; the scoped protocol and its continuum-supported frozen parameters are the construction reported here.

The retained source certificates are dated 19 September 2026; the present delivery date is 29 October 2026. Earlier work was Fable 5-led; July 2026 onward results were predominantly produced by GPT models, as reported by the operator. DeepSeek and multiple GPT models also contributed cross-review and manuscript preparation. This is reported chronology, not a per-result authorship allocation. Human names and affiliations are blank, and human expert review has not occurred. Historical source dates, the present delivery date and demonstrable public disclosure are distinct; a private commit does not establish public priority. The operator's statement is preserved verbatim:

> I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results.

References

[1] Theta=1: Full-half-line witness in finite nonlinear matter, September 2026. Retained internal mathematical source, epoch20 compact model, THETA1-CONTINUUM-PROOF.md. The model and residual argument needed here are restated in this note. Historical source date is distinct from public disclosure.

[2] Continuum trial ansatz: exact archived binary64 data, September 2026. Retained scientific input dated 19 September 2026. Exact file supplied as inputs/continuum-trial.json; SHA-256 and provenance are in PROVENANCE.json.

[3] Complete half-line interval residual partitions for the 00, 01 and 11 branches, September 2026. Retained interval certificates dated 19 September 2026. Exact files supplied as `inputs/residual-partition-00.json`, `inputs/residual-partition-01.json` and `inputs/residual-partition-11.json`. The unchanged interval producer is supplied as `reference/certify_trial_residual.py`.